Coverage and tissue geometry govern the reconstruction of spatial transcriptomics from sparse molecular measurements

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Abstract

Spatial transcriptomics reconstruction is often evaluated without distinguishing distributed missing measurements from contiguous unmeasured regions. We evaluated frozen image-only prediction, spatial interpolation and fixed image–molecular fusion across breast cancer datasets, with separate development, transportability and independent replication stages. Fresh evaluation comprised three held-out patients from one study and 22 patients from a second, using a predefined 50-gene panel. At 20% distributed measurement coverage, spatial interpolation achieved median patient Pearson correlations of 0.359 and 0.416, compared with 0.059 and 0.075 for image-only prediction. In the 22-patient replication, correlations increased from 0.380 to 0.465 as coverage rose from 10% to 40%, with lower mean squared error in every patient. Under a contiguous missing-region protocol, median correlation fell to 0.171 and error increased in all 22 patients. This contrast combines changes in geometry, scoring support and interpolation rule. Distance-related error was heterogeneous, without a universal reliability cutoff. Fixed histology fusion did not reproducibly improve interpolation; downsampled external images constrain that comparison. These findings support coverage- and geometry-aware evaluation of sparse molecular reconstruction, while identifying limits of regional completion and cross-cohort image fusion.

Keywords: spatial transcriptomics; spatial interpolation; breast cancer; missing data; external validation; histopathology

1 Introduction

Spatial transcriptomics links gene expression to positions within a tissue section, enabling molecular variation to be examined in its anatomical context[1]. Paired transcriptomic and histological resources have expanded opportunities to study relationships between tissue appearance and molecular state[2]. In breast cancer, spatial profiling has been used to characterize tumor-associated cell populations and their organization across heterogeneous tissue compartments[3–5]. These applications depend on distinguishing experimentally measured expression from values inferred at locations where molecular measurements are unavailable.

A range of computational approaches predicts spatial expression from histology. ST-Net linked local breast tumor morphology to gene expression using deep learning[6], while DeepSpaCE

examined image-based prediction and imputation in breast tissue[7]. XFuse integrates molecular measurements and images for super-resolution inference[8]. More recently, CMRCNet combined contrastive alignment with cross-modal masked reconstruction to predict expression from histopathology[9]. These methods address related, but not interchangeable, tasks: predicting expression from an image alone, fusing measured expression with images, inferring unmeasured genes, and estimating expression at unmeasured spatial locations. A benchmark of histology-to-expression methods has emphasized the importance of evaluating transportability beyond the data used for development[10].

Here we focus on a deliberately narrower task. Expression is known at a subset of locations within the same tissue section, and the aim is to reconstruct a fixed gene panel at the remaining locations. A transparent spatial interpolation baseline is particularly useful in this setting because it directly exploits the measured molecular neighborhood[11]. Comparing this baseline with an image-only predictor quantifies the value of access to same-section molecular anchors; it is not a comparison between algorithms receiving identical information. A separate comparison of interpolation with image–molecular fusion is needed to establish whether histology adds information once those anchors are available.

The spatial arrangement of missing observations also changes the prediction problem. With distributed measurements, many hidden locations lie near observed neighbors. A contiguous unmeasured region may instead require predictions to extend across a substantial gap in molecular coverage. The distinction between random and spatially structured evaluation is important whenever observations are spatially dependent[12]. Measurement fraction, nearest-anchor distance and missing-region geometry should therefore be reported explicitly rather than reduced to a single overall correlation. Error measures are also necessary: a prediction may reproduce relative spatial rankings while remaining poorly calibrated in the units used for evaluation.

We evaluated these questions through a sequence of frozen analyses spanning a four-patient development cohort, an eight-patient transportability cohort and two fresh replication studies. The final replication studies included three held-out Wu patients and 22 Bassiouni patients. We used a predefined panel of 50 genes, withheld complete expression vectors at hidden locations, and assessed distributed interpolation separately from contiguous regional completion. Supporting analyses examined graph context, image representations and sampling strategies, including negative and mixed results. The study is a multi-cohort characterization of a simple reconstruction protocol, not a proposal for a new interpolation algorithm. Its objectives are to determine how performance changes with measured coverage, to characterize spatially structured failures, and to test whether a frozen image contribution remains beneficial across cohorts.

2 Results

2.1 A staged evaluation separated development from fresh replication

The analytical design distinguished dataset roles before interpreting performance (Fig. 1; Table 1). The development cohort contained 35,536 paired locations from four HEST invasive ductal carcinoma patients, with two patients used for training, one for validation and one for

the primary development evaluation. HER2ST contributed eight patients and 3,455 locations to an initial transportability test and was subsequently reused for calibration development; it was therefore not treated as an untouched final validation cohort. The Wu study contributed six patients, of whom three were reserved for evaluation after the other three were used to establish a frozen calibration object. The second replication used the first numbered section from each of 22 Bassiouni patients, selected from source metadata before outcome evaluation. All 30,003 locations from these selected sections were assigned to evaluation, with no Bassiouni-specific training, calibration or method selection.

Fresh replication thus comprised 25 evaluation patients across two independent studies, totaling 38,484 retained locations. Both studies contained all 50 primary genes in their original order. HER2ST analyses used a 40-gene intersection and were not pooled with 50-gene summaries. The public external histology inputs were substantially downsampled: the nominal 112-μm image patch contained only 15 or 25 native pixels in the Wu cohort and 24–29 native pixels in the Bassiouni cohort. Consequently, these datasets provide direct evidence about molecular interpolation, but only a resolution-limited test of the frozen histology predictor.

Table 1: Cohorts and their analytical roles. Counts refer to retained locations and patient-level study roles, not independent spot replicates. The same eight HER2ST patients were first inspected for transportability and subsequently used for calibration development. The Wu study contains 15,601 locations in all six patients; only 8,481 locations from its three evaluation patients enter fresh validation. All 22 Bassiouni patients were evaluated without new training or calibration. "Pixels per patch" refers to the native image information in a nominal 112-μm field of view, before resizing. The primary panel was fixed in development.

Cohort	Platform	Patients, total / evaluation	Locations, total / evaluation	Genes	Analytical role and image quality
Development (HEST)	Targeted Xenium	4 / 1	35,536 / 20,761	50/50	Two training, one validation, one primary evaluation patient; conditional four-patient sensitivity analyses. Estimated 0.21–0.36 μm/pixel.
HER2ST	Sequencing ST	8 / 8	3,455 / 3,455	40/50	Initial transportability, then calibration development on the same patients. Approximately 0.69–0.92 μm/pixel.
Wu	Visium	6 / 3	15,601 / 8,481	50/50	Three calibration and three fresh evaluation patients. Only 15 or 25 native pixels per patch.
Bassiouni	Visium	22 / 22	30,003 / 30,003	50/50	First numbered section per patient; fresh replication with no cohort-specific fitting. Only 24–29 native pixels per patch.

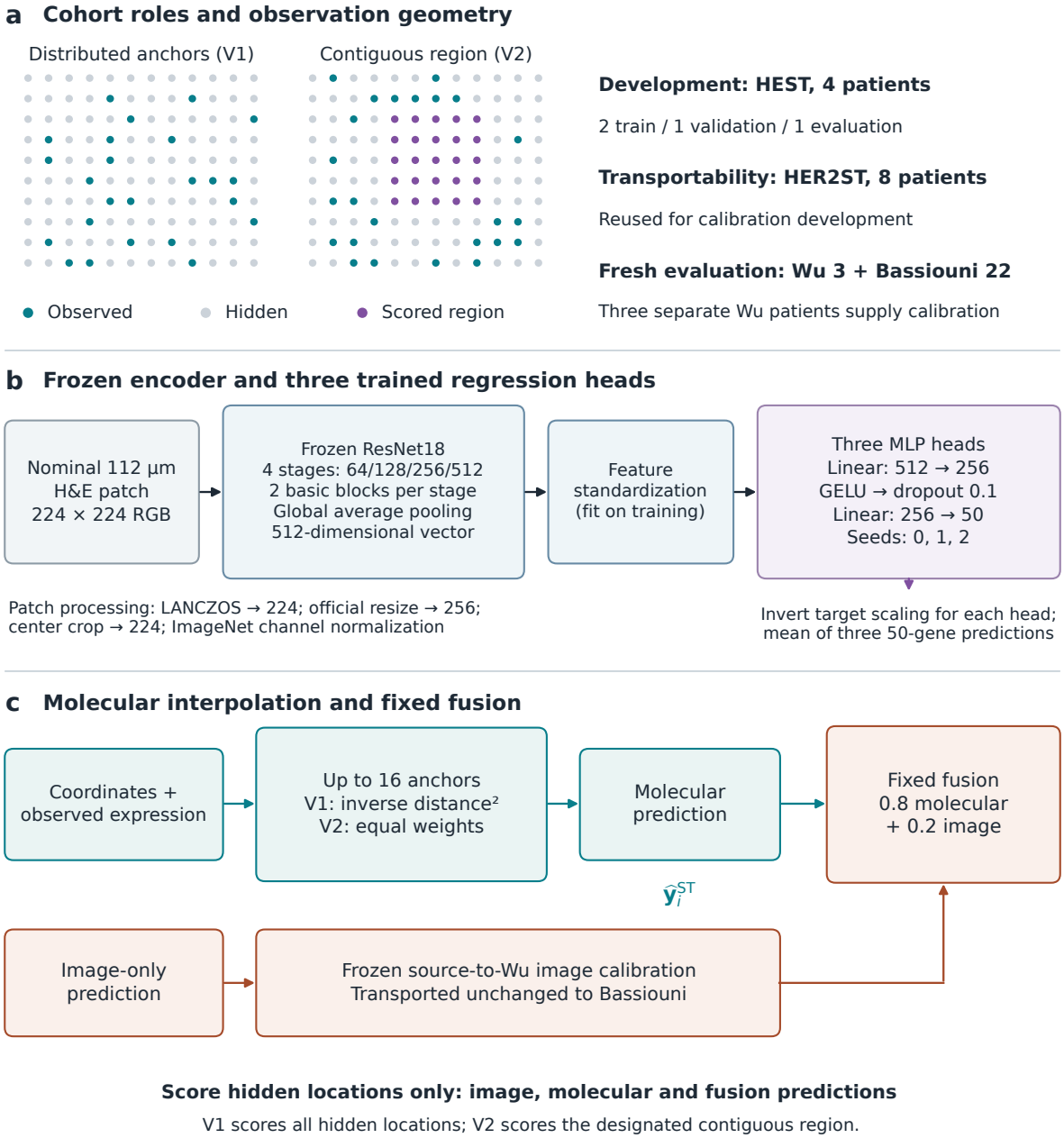


Figure 1: Study design and neural-image/molecular reconstruction architecture. a, Cohort roles and illustrative observation geometries. Teal locations are observed molecular anchors; gray locations are hidden; purple locations form the contiguous scored region. The synthetic spatial examples illustrate 20% coverage and are not patient measurements. The Wu calibration and evaluation groups contain different patients. **b,** The frozen ImageNet ResNet18 maps histology to a 512-dimensional vector after global average pooling; the ImageNet classifier is removed. External nominal 112-μm patches are first resized to 224×224 with LANCZOS, then processed by the official resize-to-256, central-crop-to-224 and channel-normalization transform. Feature scaling fitted on the training patients precedes each $512 \to 256 \to 50$ MLP, with GELU activation and dropout 0.1 after the hidden linear layer. The three heads use seeds 0, 1 and 2. Their predictions are inverse-scaled to the common target units and averaged. **c,** Molecular interpolation uses up to 16 observed neighbors, with inverse-distance-squared weighting for distributed evaluation (V1) and equal weighting for contiguous regional completion (V2). Fixed fusion combines this output with the calibrated image prediction using weights 0.8 and 0.2. The uncalibrated image ensemble is also evaluated separately. Calibration uses the development-training and Wu calibration patients and is transported unchanged to Bassiouni. Evaluation uses hidden locations only. V1 and V2 differ in weighting and scoring support as well as observation geometry.

The development comparisons represented tissue geometry explicitly through coordinate-defined graphs (Fig. 2). Each spatial location supplied a frozen image embedding, and the graph determined which neighboring embeddings contributed to its contextual representation. Neighborhood means, graph convolution and mean-GraphSAGE were evaluated using one aggregation followed by a trained expression head. Matched topology controls examined whether the resulting predictions depended on the specific physical adjacency. These comparisons informed the interpretation of spatial context; the fresh external analyses used the local image ensemble and observed-molecular reconstruction branches in Fig. 1.

2.2 Distributed molecular anchors supported reconstruction across fresh cohorts

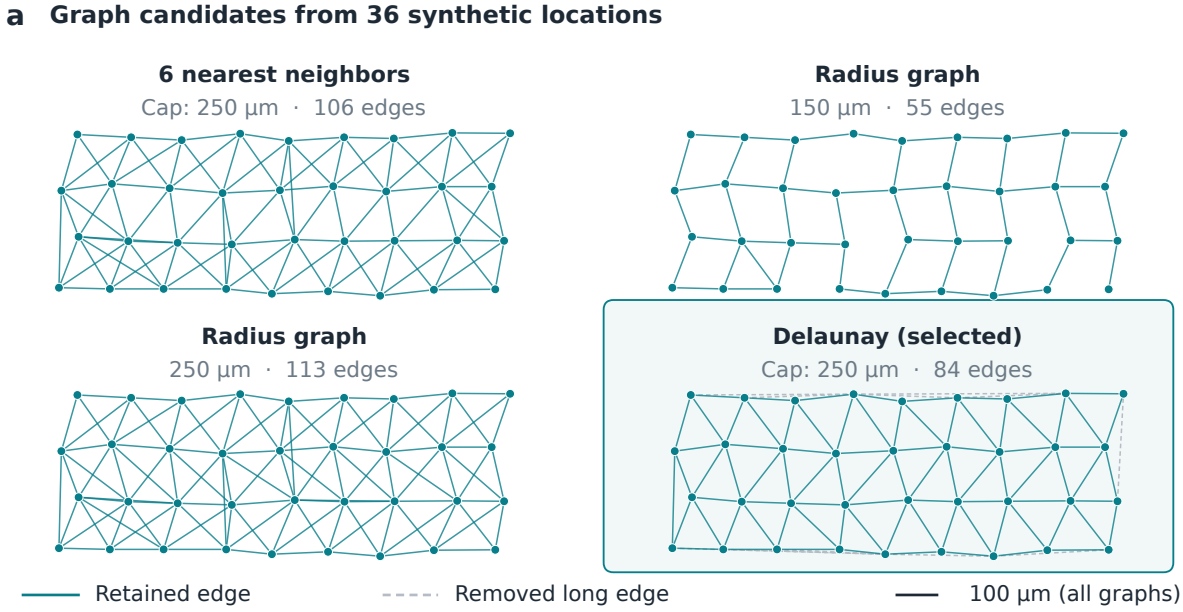
In the primary development patient, TENX99, expression at 20% distributed coverage was reconstructed with median gene Pearson correlations of 0.695 for image-only prediction, 0.792 for spatial interpolation and 0.822 for image–molecular fusion, averaged over the development masks (Fig. 3; Table 2). Mean squared errors (MSEs) were 0.911, 0.268 and 0.279, respectively. The higher fusion correlation therefore did not translate into lower error than interpolation. These values describe one development patient and do not quantify population-level generalization.

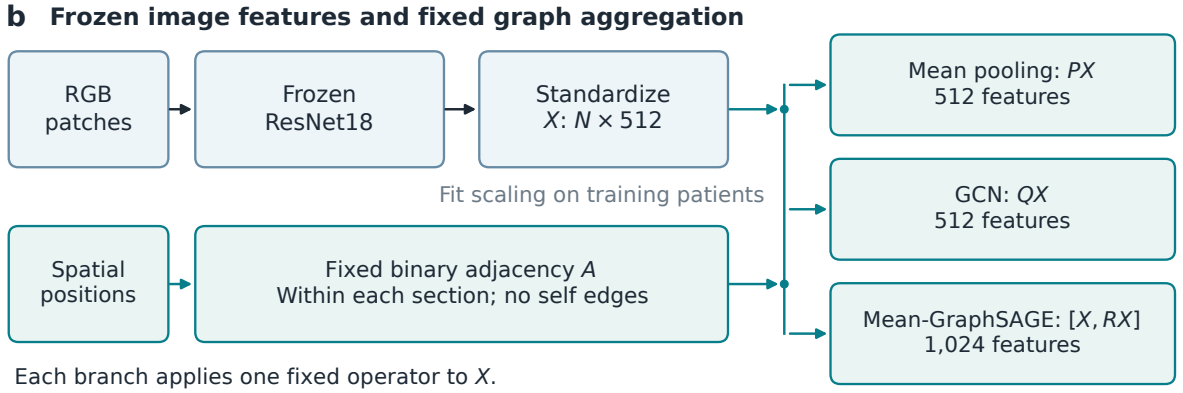
In the fresh Wu evaluation, the median of patient-level median gene correlations at 20% distributed coverage was 0.359 for spatial interpolation and 0.059 for image-only prediction (Fig. 4). Corresponding MSEs were 0.0987 and 2.9155, and mean absolute errors (MAEs) were 0.2179 and 1.5578. All three evaluation patients favored interpolation over the frozen image predictor in correlation, MSE and MAE at every tested budget. The large error reduction is partly relative to an image predictor with substantial cross-platform scale and image-resolution mismatch; it should not be interpreted as accurate recovery of absolute transcript abundance or superiority over all image-based methods.

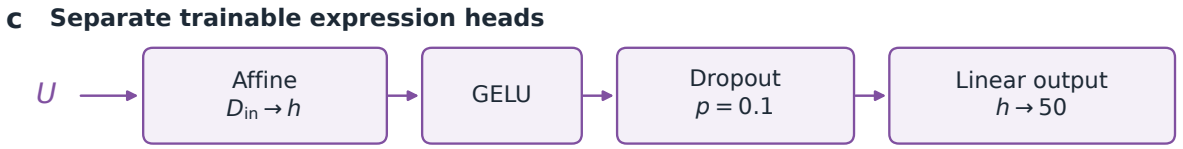
The second fresh replication provided consistent support for the same comparison (Fig. 5). At 20% distributed coverage, spatial interpolation achieved a median patient correlation of 0.416, MSE of 0.0923 and MAE of 0.2033. Image-only values were 0.075, 1.9673 and 1.2120. All 22 patients met the prespecified joint support criterion at each of the three budgets: a correlation gain of at least 0.02, relative MSE reduction of at least 2%, and non-worsening MAE. This is a descriptive support criterion, not a significance test. Because interpolation receives measured expression from the evaluation section whereas the image-only model does not, the result establishes the utility of molecular anchors under the specified task rather than an equal-information algorithmic comparison.

2.3 Increasing distributed coverage improved reconstruction consistently

Increasing the fraction of observed locations improved distributed reconstruction in both fresh cohorts (Supplementary Table S9). Across the Bassiouni patients, median patient correlations at 10%, 20% and 40% coverage were 0.380, 0.416 and 0.465; MSEs were 0.1001, 0.0923 and 0.0852. Every patient showed higher correlation and lower MSE at 40% than at 10%. The Wu cohort showed the same directional pattern in all three patients, with correlations of 0.313, 0.359 and 0.393 and MSEs of 0.1080, 0.0987 and 0.0899.







Mean / GCN:	512 → 128 → 50	72,114 parameters
GraphSAGE:	1,024 → 67 → 50	72,075 parameters
Local (no graph):	512 → 128 → 50	72,114 parameters

Development comparisons; topology-specific gains were not established.

Figure 2: Coordinate-defined graph context in development predictors. a, All retained edges are shown for the same 36 synthetic locations: the symmetrized union of six-nearest-neighbor edges capped at 250 μm, radius graphs at 150 and 250 μm, and Delaunay edges capped at 250 μm. Every graph has the same physical scale. Gray dashed lines mark Delaunay edges removed by the cap. Symmetrization can yield more than six neighbors. Capped Delaunay was selected on validation data. These locations are illustrative, not patient data. **b,** Frozen ResNet18 features are standardized using training-patient statistics. Coordinate-defined binary adjacency supplies one fixed aggregation, as defined in Eq. 3. The three branches are alternatives, not successive layers. **c,** Separate trainable heads map each alternative representation to 50 gene outputs; inverse target scaling restores common expression units. The local comparator uses unaggregated image features. Both affine layers include biases; dropout is disabled at inference. These initial development comparisons are separate from the final external prediction pipeline. Matched controls did not establish a reproducible topology-specific gain (Supplementary Fig. S5).

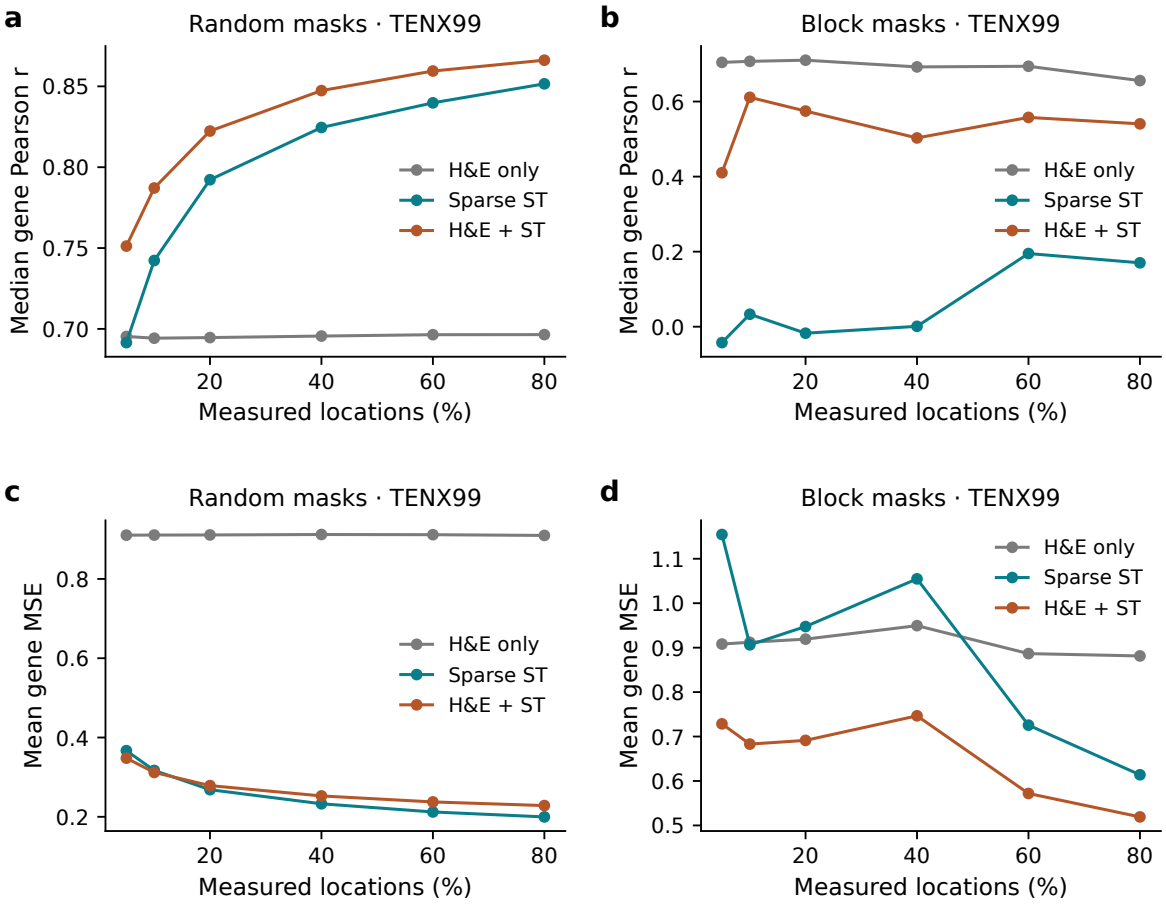


Figure 3: Development reconstruction depends on molecular coverage and masking regime. Results are from the single primary development patient, TENX99, using the frozen 50-gene panel and ResNet18 representation. **a,c,** Distributed random observation. **b,d,** The earlier development block-mask regime. Panels a and b show median gene Pearson correlation; panels c and d show mean gene MSE across measurement fractions. The image-only, selected molecular interpolation and fixed fusion curves summarize three frozen mask replicates per condition; curves are means over masks, not patient confidence intervals. Targets and error units are unchanged across methods. The development block masks differ from the external V2 forbidden-region protocol and should not be treated as directly equivalent geometry tests. At 20% distributed coverage, fusion has higher correlation than interpolation but slightly higher MSE, illustrating that spatial association and calibration are distinct properties. The image-only comparator receives no observed molecular anchors.

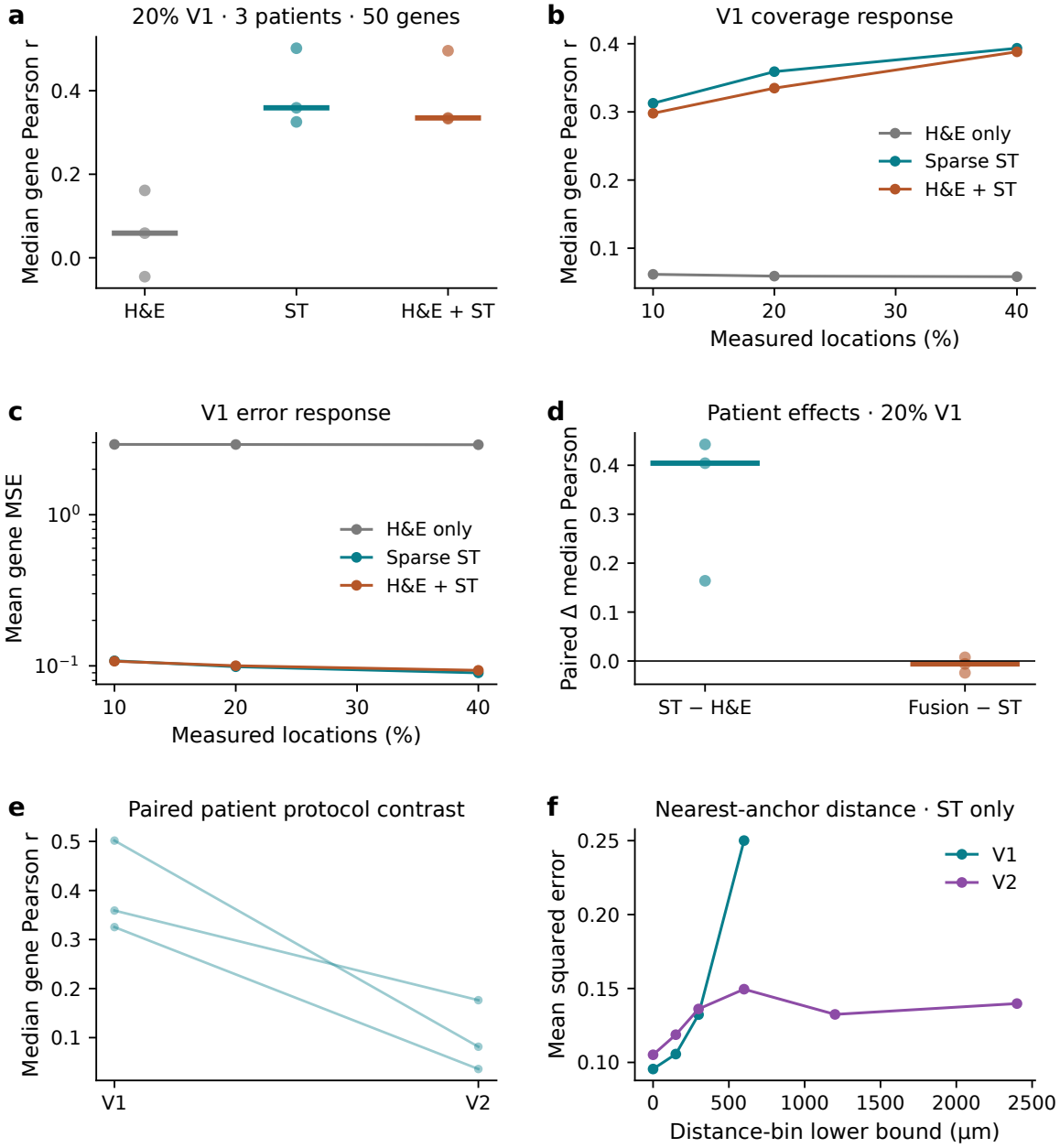


Figure 4: Fresh evaluation in three held-out Wu patients supports interpolation but not robust image fusion. The frozen 50-gene panel was evaluated in patients 1160920F, CID4290 and CID44971, none of whom contributed to Wu calibration. **a,** Patient-level correlation summaries at 20% distributed coverage. **b,c,** Median patient correlation and MSE across 10%, 20% and 40% distributed coverage. **d,** Paired patient correlation effects for interpolation versus image-only prediction and fusion versus interpolation. **e,** Paired distributed (V1) and contiguous-region (V2) molecular correlations at 20%. **f,** Distance-resolved molecular MSE at 20%. Patient summaries average ten random mask seeds within geometry and then geometry means; cohort summaries are medians, not pooled spot estimates. Three fixed contiguous geometries are used for V2. Each patient is a biological replicate; seeds are not. V1 uses inverse-distance-squared interpolation and V2 uses a nearest-anchor mean, so panel e is a protocol contrast. The public image inputs contained only 15 or 25 native pixels per 112-μm patch, limiting conclusions about morphology. Error units are common development-standardized log1p counts.

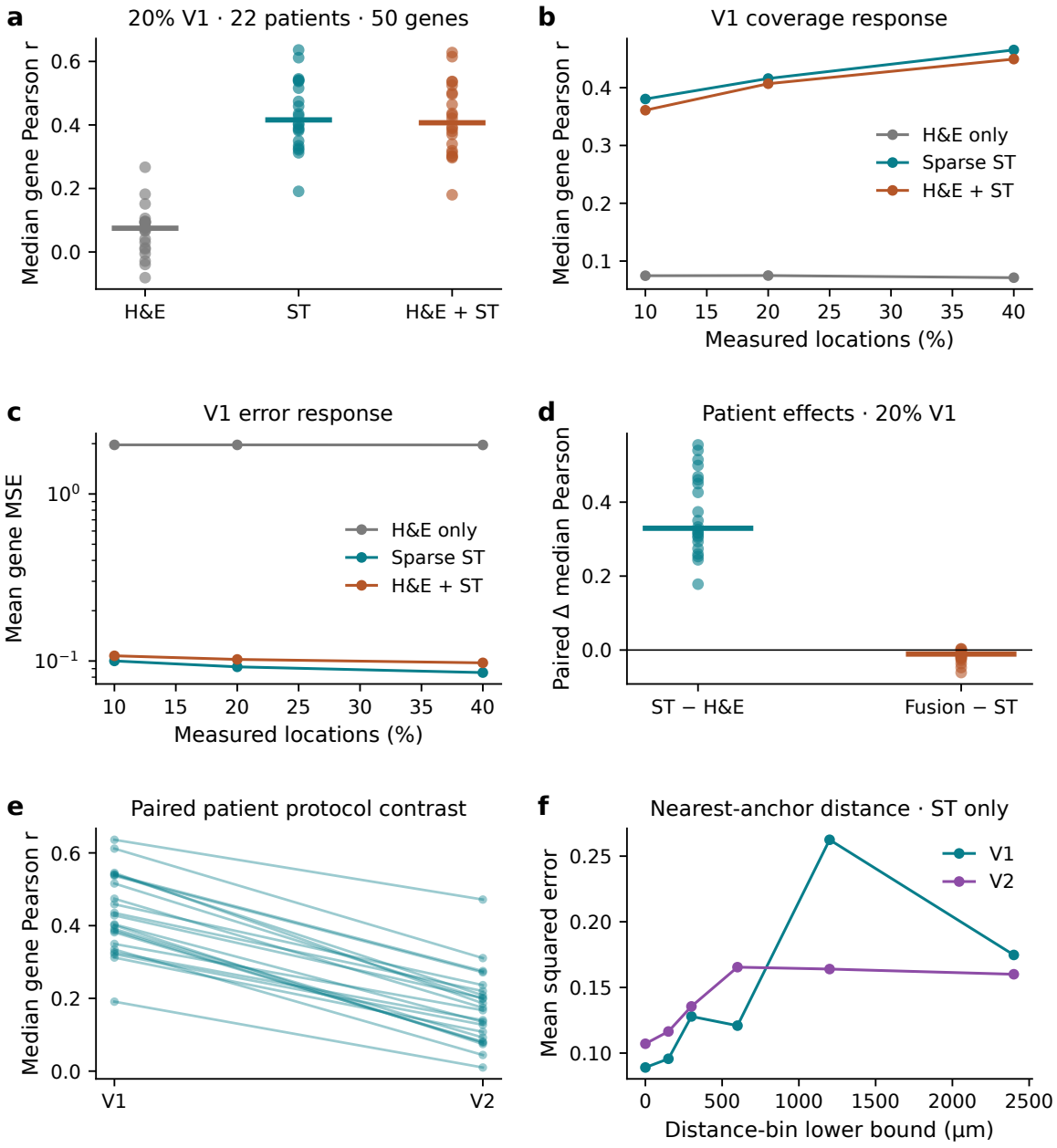


Figure 5: Independent 22-patient replication confirms the coverage response and weaker contiguous completion. One first-numbered section per Bassiouni patient was selected before evaluation, yielding 30,003 locations and the same 50 primary genes. No patient from this study was used for new training or calibration. **a,** Patient correlation summaries at 20% distributed coverage. **b,c,** Correlation and MSE across measurement fractions. **d,** Paired interpolation–image and fusion–interpolation correlation effects. **e,** Paired V1/V2 molecular correlations at 20%. **f,** Distance-resolved molecular MSE at 20%. Each patient summary averages ten mask seeds within geometry and then the three V2 geometry means where applicable; cohort curves use patient medians. All 22 patients improved in correlation and MSE between 10% and 40% V1 coverage and deteriorated in correlation, MSE and MAE under V2 relative to V1 at 20%. V1/V2 differ in scoring support and interpolation weighting as well as missing-region design. Image fusion uses the frozen Wu calibration and 0.2 image weight. Public histology contained only 24–29 native pixels per 112-μm patch. Distance curves have varying occupied-bin support and are not calibrated uncertainty intervals. No p-values or patient confidence intervals are implied.

Table 2: Reconstruction at 20% distributed measurement coverage. Each entry is a cohort median of patient-level summaries after averaging mask seeds within geometry and geometries within patient, except that the development row represents the single primary patient. Pearson r is the median correlation across panel genes; MSE and MAE are patient means across genes and hidden locations before the cohort median. All methods within a cohort use identical target units and observation masks. Different gene panels and assay count distributions preclude treating these values as a pooled cross-cohort ranking. Fusion rules and their calibration are frozen but differ between the development and external protocols; see Methods and Supplementary Information. The image-only branch has no same-section molecular anchors.

Cohort	n	Genes	Method	Pearson r	MSE	MAE
Development	1	50	Image only	0.695	0.9110	0.7606
			Spatial interpolation	0.792	0.2684	0.3904
			Fixed fusion	0.822	0.2787	0.4173
HER2ST, zero-shot	8	40	Image only	0.143	4.3665	1.9376
			Spatial interpolation	0.194	0.0680	0.1730
			Fixed fusion	0.255	0.3481	0.5153
Wu, fresh	3	50	Image only	0.059	2.9155	1.5578
			Spatial interpolation	0.359	0.0987	0.2179
			Fixed fusion	0.335	0.1002	0.2321
Bassiouni, fresh	22	50	Image only	0.075	1.9673	1.2120
			Spatial interpolation	0.416	0.0923	0.2033
			Fixed fusion	0.407	0.1022	0.2288

These curves describe the tested measurement fractions and do not identify a universally sufficient sampling budget. Coverage denotes the fraction of retained spatial locations whose complete expression vectors are revealed, not the fraction of tissue area assayed or a proportional reduction in experimental cost. Random sampling used nested anchor sets, but the hidden evaluation set also changes with coverage. The observed improvements therefore apply to the frozen budget-specific reconstruction protocols rather than to an identical set of targets at every budget. Label-free spatial coverage sampling was retained as a secondary analysis; earlier sampling comparisons had not established a reproducible advantage for difficult contiguous missing regions (Supplementary Fig. S8).

2.4 Contiguous regional completion exposed substantially weaker spatial prediction

Under the contiguous missing-region protocol, reconstruction quality deteriorated across the fresh cohorts (Figs. 4 and 5; Supplementary Table S9). At 20% coverage in the Bassiouni cohort, median patient correlation decreased from 0.416 for distributed interpolation to 0.171 for regional completion, while MSE increased from 0.0923 to 0.1496 and MAE from 0.2033 to 0.2624. All 22 patients deteriorated in all three measures. The median paired correlation change was -0.267, and the median paired relative MSE increase was 52.8%. The latter is not the ratio calculated from separate cohort medians.

The Wu cohort showed a similar limitation: median patient correlation under the contiguous protocol was 0.0815, compared with 0.359 for distributed interpolation. Every patient had higher error and lower correlation. These results show that a favorable distributed-masking

score does not imply equally reliable recovery across a large unmeasured region.

This contrast must nevertheless be interpreted at the protocol level. Distributed evaluation used inverse-distance-squared interpolation and scored all hidden locations; regional evaluation used a nearest-anchor mean and scored the designated contiguous region only. Thus the comparison changes the interpolation rule, the spatial arrangement of observed locations and the composition of the scored tissue. It does not isolate a causal effect of geometry. Moreover, a withheld region can lie inside the convex hull of observed anchors. We use "contiguous extrapolation" only as an operational description of extending predictions into a region without molecular observations, not as proof of geometric out-of-support extrapolation.

2.5 Anchor distance was informative but did not define a universal reliability cutoff

Distance-resolved error summaries complemented the panel-level results and spatial error maps (Fig. 6). In the Bassiouni cohort at 20% coverage, the farthest occupied distance bin had higher MSE than the nearest bin in 15 of 22 patients for distributed interpolation and 20 of 22 for contiguous regional completion. Strictly increasing occupied-bin MSE curves occurred in only 6 and 13 patients, respectively. The Wu distributed curves increased across occupied bins in all three patients, whereas regional curves were nonmonotonic.

These heterogeneous patterns do not support a single extrapolation horizon applicable across patients, genes and missing-region designs. Distance bins contain different tissue locations and may differ in expression variance and tissue composition; remote bins can also contain relatively few observations. We therefore report bin support, omit empty bins and retain spatial error maps rather than assigning a universal distance cutoff (Supplementary Table S7). The analysis describes an association between distance and reconstruction error without controlling all other sources of spatial heterogeneity.

2.6 Frozen image fusion did not provide a reproducible external advantage

The benefit of access to molecular anchors did not imply a benefit from adding the tested image prediction to those anchors. In the Bassiouni cohort at 20% distributed coverage, fixed fusion yielded a median patient correlation of 0.407, MSE of 0.1022 and MAE of 0.2288, compared with 0.416, 0.0923 and 0.2033 for interpolation alone. Fusion increased correlation in only 3 of 22 patients and improved neither MSE nor MAE in any patient. Median paired correlation change was -0.0108, with a median paired relative MSE increase of 10.7%. In the Wu cohort, only one of three patients improved in correlation and none improved in either error measure.

HER2ST transportability analyses had already exposed a substantial image/fusion calibration problem (Supplementary Figs. S1 and S9). Recalibration during subsequent development reduced scale-related error, but selected fusion rules did not establish robust joint correlation and error improvements (Supplementary Fig. S10). The final replication deliberately retained the Wu-derived calibration object and the fixed 0.2 image weight rather than adapting to Bassiouni outcomes. Together with external image downsampling, this restricts the conclusion to the frozen models and available inputs; it does not establish that high-resolution histology cannot improve a differently designed reconstruction method.

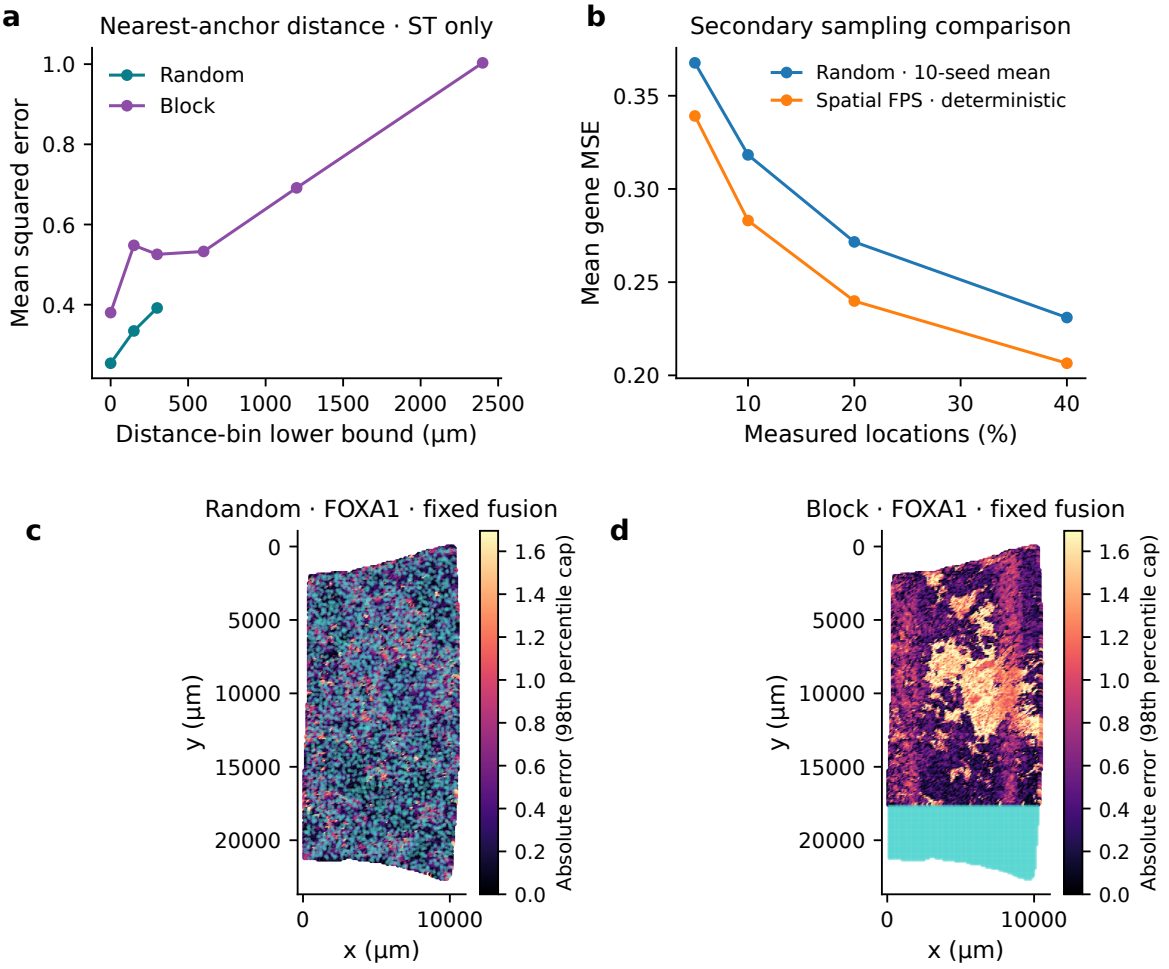


Figure 6: Spatial diagnostics reveal heterogeneous error beyond an overall prediction score. Development results use TENX99 and the frozen 50-gene panel. **a,** Mean gene MSE versus nearest-anchor distance for molecular interpolation at 20% coverage under distributed and development block masks; only occupied distance bins are shown. **b,** Development comparison of random and spatial farthest-point sampling, shown as MSE at matched measured-location budgets. These are descriptive single-patient sampling diagnostics, not a demonstration of universally optimal acquisition. **c,d,** Absolute-error maps for the preselected gene *FOXA1*, fixed fusion, 20% coverage and replicate 0 under random and block masks. Both maps use the same error scale, capped at the 98th percentile of the pooled preselected map-source values, which also include the archived capture-window condition; no locations are removed. Missing-region shape, available anchors and scored tissue differ between the maps. The figure illustrates why distance-bin support and spatial errors should accompany global correlation and why development geometry findings require independent evaluation.

A separate invariance check clarified what target scaling could establish. Per-gene affine transformation followed by normalized linear interpolation and inverse transformation is algebraically equivalent to interpolation in the original target space. The molecular "harmonized" and unharmonized predictions consequently agreed to numerical precision. We did not count them as distinct algorithms or attribute improved molecular interpolation to this transformation (Supplementary Note: affine interpolation identity).

2.7 Controls and provenance audits bounded the interpretation of the results

Anchor-expression and anchor-position assignment permutations degraded molecular reconstruction in the predefined control cases, supporting dependence on the correct relationship between measured expression and location. These controls are related disruptions of the same association rather than independent mechanistic tests. Image permutation did not provide comparable evidence for a useful image contribution (Supplementary Figs. S11 and S12; Supplementary Table S4). Earlier morphology, graph-topology and encoder experiments were retained transparently and did not justify claims that exact graph topology was essential (Supplementary Figs. S3–S7).

The second replication produced 2,904 clean and 18 control bundles, containing 8,766 method predictions. All 17,532 hidden-label NaN/random corruption comparisons were invariant. The recorded audit reproduced 8,766 predictions and 35,064 per-gene metric arrays exactly and reconstructed 968 label-free mask trajectories. These checks establish computational traceability within the frozen workflow; they do not replace independent biological replication or remove limitations of the target representation (Supplementary Tables S5 and S6).

3 Discussion

This study shows that a simple, frozen spatial interpolation protocol can recover part of the variation in a predefined expression panel when molecular measurements are available at distributed locations in the same tissue section. The direction of benefit over the frozen image-only comparator was consistent in two fresh breast cancer studies, including all 22 patients in the larger replication. Increasing measured-location coverage improved correlation and reduced error, whereas contiguous regional completion remained substantially weaker. The contribution is a characterization of reconstruction reliability under explicit information and masking constraints, not a new interpolation algorithm or a claim of transcriptome-wide recovery.

The comparison with histology requires particular care. Molecular interpolation uses patient-specific observations from the evaluation section and therefore has access to information unavailable to the image-only model. Its advantage demonstrates the practical value of these anchors but does not establish superiority over all histology-based prediction methods. Prior image-based and multimodal approaches address different training and inference settings[6–9]. The appropriate test of incremental morphology in our setting was fusion versus interpolation. That test did not show a reproducible benefit. Low-resolution public images, a generic ImageNet representation and transported calibration constrain the comparison; a comprehensive

benchmark of native-resolution pathology models was outside the frozen replication protocol.

The contiguous missing-region results make the evaluation design as important as the average score. A method that interpolates between nearby measurements can perform poorly when observations are absent from a substantial tissue region. This is consistent with the broader need to match validation structure to the intended prediction task in spatially dependent data[12]. However, our V1/V2 protocols also differ in interpolation family and scored locations. A future factorial study holding the interpolation rule and evaluation support fixed would be needed to estimate the isolated contribution of missing-region geometry. In the present analysis, the defensible conclusion is that the entire contiguous-completion protocol is harder, not that geometry alone explains the measured decline.

Distance-to-anchor summaries expose additional heterogeneity but are not calibrated uncertainty estimates. The absence of a universal monotonic curve suggests that distance alone is insufficient to determine where a reconstructed expression value is trustworthy. Reconstruction may also be affected by variation in the target field and the composition of the scored locations. Uncertainty-aware approaches such as TISSUE illustrate why uncertainty in predicted expression matters for downstream analyses[13]; we did not implement that framework or claim calibrated intervals here. Regional errors, low-support bins and patient-specific failures should remain visible before reconstructed values are used for differential expression, cell-state interpretation or other biological analyses.

Cross-platform scale differences present a distinct challenge. A common numerical transformation does not make targeted in situ counts and sequencing-based capture-spot counts biologically interchangeable. Capture support, depth and composition remain different even after values are centered and scaled. The observed reduction in image/fusion error during HER2ST calibration development should therefore not be confused with a validated improvement in molecular interpolation. The affine identity shows why the latter comparison is unidentifiable for the specified normalized weighted average. It also provides a useful implementation check: negligible numerical rank differences should not be interpreted as biological gains.

Several limitations define the scope of the study. The primary panel was selected in development and contained 50 genes, with only 40 shared in HER2ST. The results do not establish recovery of unmeasured genes or an entire transcriptome. Missingness was simulated by withholding existing measurements, not created through a prospective reduced-assay experiment. Reported coverage is a proportion of retained locations and does not translate directly to tissue area, sequencing expenditure or financial savings. Patient numbers were small in the first fresh validation, and all cohorts were breast tissue. External images were downsampled and the frozen fusion rule was not optimized for them. We also did not benchmark a comprehensive set of spatial statistical or modern histology-to-expression methods under equal information access. Consequently, the current evidence should not be read as state-of-the-art superiority or clinical utility.

The sequential design offers a complementary strength. Development, inspected transportability cohorts and fresh evaluation cohorts were kept distinct, and mixed results were retained rather than used to retroactively revise the external protocol. Technical replay, mask provenance and hidden-label corruption checks reduce the risk that an apparent reconstruction benefit arises from leakage. These safeguards are especially important when measurement masks expose expression from the evaluation tissue itself. Nevertheless, the decision rules

and repeated development analyses remain part of an iterative research process rather than a substitute for prospective experimental confirmation.

Overall, sparse molecular anchors provided useful reconstruction under distributed sampling, with a consistent coverage response across fresh cohorts. The same evidence delineated weaker contiguous regional completion and a lack of reproducible benefit from the tested image fusion. Reporting measurement coverage, missing-region design, anchor distance, calibration behavior and patient-level effects together gives a more informative account of reliability than a single pooled prediction score.

4 Methods

4.1 Reconstruction framework

The reconstruction task estimates a fixed panel of G expression targets at unobserved spatial locations. For a section with N retained locations, each location i has physical coordinates p_i, an image patch I_i and a target vector $\boldsymbol{y}_i$ in the common units defined below. The observed set $\mathcal{O}$ supplies complete expression vectors; coordinates and image patches are available at every retained location. Predictions are scored only where expression was withheld.

The final reconstruction framework combined a frozen neural image encoder and trained expression head with a separate molecular interpolation branch (Fig. 1). Both branches predicted the same ordered 50-gene panel. Molecular interpolation used spatial coordinates and observed expression; final fusion used fixed molecular and calibrated-image weights of 0.8 and 0.2. The graph neural comparisons in Fig. 2 assessed contextual image representations during development. Their architecture, spatial operators and supervision are defined separately below.

4.2 Neural image encoder and local representation learning

The frozen ResNet18[14] comprised a 7×7 convolutional stem with stride 2, batch normalization, a rectified-linear activation and 3×3 max pooling with stride 2, followed by four residual stages. The stages contained two basic blocks each, with channel widths 64, 128, 256 and 512. Each block used two 3×3 convolutions with batch normalization and a rectified-linear activation after the first convolution and after shortcut addition. Identity or projected shortcuts matched the feature dimensions. Global average pooling produced a 512-dimensional vector; the original ImageNet classifier was replaced by the identity operation. All encoder parameters and batch-normalization statistics remained fixed during feature extraction.

The selected local expression head contained one hidden layer with the sequence

$$512 \xrightarrow{\text{linear}} 256 \xrightarrow{\text{GELU}} 256 \xrightarrow{\text{dropout (0.1)}} 256 \xrightarrow{\text{linear}} 50, \tag{1}$$

where GELU denotes the Gaussian error linear unit. Both linear layers included biases, giving 144,178 trainable parameters. The output was linear and unconstrained. The head learned a per-location representation from the frozen image embedding under supervised expression regression.

Let S_x denote feature-wise standardization and S_y the additional head-target standardization, both fitted on the training patients. If f_{θ_s} denotes the head trained with seed $s \in \{0, 1, 2\}$ and ϕ_{R18} the frozen encoder, the image prediction in the common target units was

$$\widehat{\boldsymbol{y}}_i^{\mathrm{image}} = \frac{1}{3} \sum_{s=0}^{2} S_y^{-1}[f_{\theta_s}(S_x(\phi_{\mathrm{R18}}(I_i)))] \,. \tag{2}$$

Dropout was disabled at inference. The three inverse-scaled predictions were averaged with equal weights. Image preprocessing, scaler estimation, optimization and validation selection are specified under Experimental setup.

4.3 Spatial graph construction

The graph comparisons represented each tissue section as a separate undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. A node $i \in \mathcal{V}$ denoted one retained spatial-transcriptomic location, with registered physical coordinates $p_i \in \mathbb{R}^2$ in micrometers and a corresponding histology patch. Its initial feature vector $x_i \in \mathbb{R}^{512}$ was the frozen ResNet18 embedding after feature-wise standardization fitted on the training patients (Fig. 2). The rows of $X \in \mathbb{R}^{N \times 512}$ followed the same barcode order as the coordinates and expression targets. Graph construction used coordinates only; neither expression values nor histology-derived similarities determined the edges. Graphs never connected different sections. At evaluation, the graph could use the coordinates and image embeddings of all retained locations in the section, but evaluation expression entered only the subsequent scoring operation.

We considered three geometric constructions. For the nearest-neighbor graph, each node nominated up to its six closest other locations within 250 μm. Equal-distance candidates were ordered by their node index. The undirected edge set was the union of these nominations, so a node's final degree could exceed six. Radius graphs connected every pair with Euclidean separation at most r, for $r = 150$ or 250 μm. For the Delaunay graph, we triangulated the physical coordinates, collected the three sides of each triangle, and removed edges longer than 250 μm. Delaunay triangulation uses the empty-circumcircle construction[15]; it was computed with the SciPy implementation[16]. The distance cap restricted long connections across gaps in the sampled coordinates. It did not encode a segmented tissue boundary or a biological interaction.

Duplicate undirected edges were removed, and the binary adjacency $A \in \{0, 1\}^{N \times N}$ satisfied $A = A^\top$ and $A_{ii} = 0$. We write $\mathcal{N}(i) = \{j : A_{ij} = 1\}$ and $\delta_i = |\mathcal{N}(i)|$. Self-connections were introduced by the aggregation operator where required, rather than being stored as edges in A. Graph choice and the numerical settings selected for each predictor are reported in Experimental setup.

4.4 One-step graph aggregation and learned node representations

Message-passing models form node representations by combining information from adjacent nodes with a learned update[17]. Our contextual predictors used one such spatial aggregation followed by a learned hidden representation and a node-wise expression readout. The physical edge rule determines which neighboring image embeddings can contribute to each

location. The adjacency and aggregation coefficients were fixed; learning operated on the feature-channel transformations. We compared a self-inclusive neighborhood mean, a symmetrically normalized graph convolution[18], and a mean-GraphSAGE adaptation that retained the central-node vector separately from its neighbors[19].

Define $\widetilde{A} = A + I$, $\widetilde{D} = \mathrm{diag}(\delta_i + 1)$ and $D_\star = \mathrm{diag}(\max(\delta_i, 1))$. The operators were

$$P = \widetilde{D}^{-1}\widetilde{A}, \qquad Q = \widetilde{D}^{-1/2}\widetilde{A}\widetilde{D}^{-1/2}, \qquad R = D_\star^{-1}A. \tag{3}$$

In the graph equations, $x_i = X_{i,:}$, m_i and $h_i = \mathsf{H}_{i,:}$ are row vectors, and $\mathbf{1}$ is an N-dimensional column of ones. The corresponding messages arriving at location i were

$$m_i^{\mathrm{mean}} = \frac{x_i + \sum_{j\in\mathcal{N}(i)} x_j}{\delta_i + 1}, \tag{4}$$

$$m_i^{\mathrm{GCN}} = \sum_{j\in\mathcal{N}(i)\cup\{i\}} \frac{x_j}{\sqrt{(\delta_i+1)(\delta_j+1)}}, \tag{5}$$

$$m_i^{\mathrm{SAGE}} = \frac{1}{\max(\delta_i, 1)} \sum_{j\in\mathcal{N}(i)} x_j. \tag{6}$$

Thus P averages the central location and its neighbors, while Q adjusts each contribution using both endpoint degrees and is generally not a row-normalized mean. The GraphSAGE message uses neighbors only; an isolated node receives the zero neighbor vector and retains its own vector through the separate central-node input. In the mean and GCN models, an isolated node's self-connection preserves its own feature vector.

The input matrix supplied to the learned head was

$$U = \begin{cases} PX, & \text{neighborhood mean,} \\ QX, & \text{GCN,} \\ [X \,\|\, RX], & \text{mean-GraphSAGE,} \end{cases} \tag{7}$$

where $\|$ denotes feature concatenation. The hidden node representation and prediction were

$$\mathsf{H} = \mathrm{GELU}(UW_1 + \mathbf{1}b_1^\top), \tag{8}$$

$$\widehat{Y} = S_y^{-1}\left[\mathrm{Dropout}_p(\mathsf{H})W_2 + \mathbf{1}b_2^\top\right], \tag{9}$$

with $W_1 \in \mathbb{R}^{D_{\mathrm{in}}\times h}$, $b_1 \in \mathbb{R}^h$, $W_2 \in \mathbb{R}^{h\times G}$ and $b_2 \in \mathbb{R}^G$. Here G is the number of genes in the fixed panel, and S_y^{-1} reverses the additional target standardization fitted on training patients. The output has one unconstrained real value per location and gene in the common expression-target units. Dropout operates after GELU during training and becomes the identity at inference.

For the selected 50-gene ResNet18 comparison, the mean and GCN predictors used $D_{\mathrm{in}} = 512$, $h = 128$ and $p = 0.1$, giving 72,114 learned parameters. The mean-GraphSAGE input concatenated two 512-dimensional vectors and used $D_{\mathrm{in}} = 1024$, $h = 67$ and $p = 0.1$, giving 72,075 learned parameters. Its smaller hidden width approximately matched the parameter count of the initial local $512 \to 128 \to 50$ predictor. It was not the width-256 local head subsequently selected for the representation comparison and reused in the final external image ensemble.

The GraphSAGE first-layer matrix can be partitioned into central and neighbor blocks, yielding the equivalent node update

$$h_i = \mathrm{GELU}\Big(x_i W_{\mathrm{self}} + m_i^{\mathrm{SAGE}} W_{\mathrm{neigh}} + b_1^{\top}\Big). \tag{10}$$

The two learned blocks allow central morphology and surrounding morphology to contribute differently. For GCN, $(QX)W_1 = Q(XW_1)$, so precomputing QX is algebraically equivalent to the linear propagation component of a single graph-convolutional layer before its nonlinearity[18]. The same factorization applies to the neighborhood mean. This implementation aggregates the complete fixed neighborhood once; it does not add another graph operator after the hidden layer or use repeated graph propagation, learned edge weights, attention, stochastic neighbor subsampling or post-update ℓ_2 normalization. These details distinguish the implemented one-step adaptations from the full training architectures of the cited methods.

Only W_1, b_1, W_2, b_2 were optimized using supervised expression targets. The encoder, coordinates, graph, degree normalization and feature scaler remained fixed. The rows of H are therefore learned contextual representations of individual locations, with a one-hop graph receptive field, rather than a learned embedding of the entire tissue graph. The output readout shares parameters across locations and performs no graph-level pooling. No contrastive, graph-reconstruction or neighborhood-expression objective was used. Optimization and validation selection are specified in Experimental setup.

4.5 Topology controls and representation comparison

The topology controls separated access to additional morphology from dependence on particular spatial edges. A self-only control supplied $[X\|X]$ to the same GraphSAGE head: this preserves its nominal parameter count, but the two inputs contain redundant local information. A vertex-permutation control reassigned the graph's vertices while leaving image features in their physical order; it preserved the abstract graph and degree distribution while disrupting the alignment between adjacency and tissue location. Degree-preserving double-edge swaps retained each location's degree. A further constrained version also retained the global counts of edges in prespecified physical length intervals. The swap schedule, length bins and replicate structure are given in Experimental setup.

These controls address different sources of contextual benefit. Similar performance after constrained rewiring does not establish that all spatial information is dispensable, because the constrained graphs retain local connections and many original edges. Parameter matching likewise does not make the self-only and contextual function classes identical. We therefore interpreted the observed mixed control results as insufficient evidence for a reproducible advantage attributable to exact adjacency, without claiming that graph topology is universally uninformative.

To examine dependence on the image representation, the frozen Phikon ViT-B/16 encoder[20] used the public `owkin/phikon` model pinned to revision `057cc0295895`. Its 12-layer transformer supplied the final-layer-normalized 768-dimensional classification-token vector, without a pooler. The local 50-gene head had dimensions $768 \to 256 \to 50$ and 209,714 trainable parameters. Supplementary Table S8 summarizes the selected local and contextual architec-

tures. In the spatial sensitivity comparison, the same graph operators acted on standardized 768-dimensional features; the mean predictor used $768 \to 128 \to 50$ and mean-GraphSAGE used $1536 \to 67 \to 50$. The inherited widths and adjacency isolate some modeling choices but do not match parameter counts across encoders. The graph and alternative-representation analyses were development comparisons; the fresh external reconstruction used the frozen local ResNet18 ensemble and the separately defined observed-molecular interpolation and fixed-fusion rules.

4.6 Spatial interpolation and image–molecular fusion

The molecular predictor used the up to 16 nearest observed anchors, $\mathcal{N}_{16}(i;\mathcal{O})$. For distinct observed and hidden coordinates, distributed interpolation used normalized inverse-distance-squared weights, a local form of inverse-distance interpolation[11]:

$$\widehat{y}_{ig}^{\mathrm{ST}} = \sum_{j\in\mathcal{N}_{16}(i;\mathcal{O})} w_{ij} y_{jg}, \qquad w_{ij} = \frac{\|p_i - p_j\|^{-2}}{\sum_{k\in\mathcal{N}_{16}(i;\mathcal{O})} \|p_i - p_k\|^{-2}}. \tag{11}$$

For the frozen V2 protocol, the same neighborhood size was used with equal weights $w_{ij} = 1/|\mathcal{N}_{16}|$. These rules were fixed during the Wu evaluation and were not selected on Bassiouni outcomes. Fewer than 16 anchors, if available, were handled by using all available anchors. Euclidean distances were rounded to six decimal places in micrometers and floored at 10^{-6} μm before inverse-square weighting; the normalized weights therefore remained finite at coincident coordinates.

The image-only prediction was denoted $\widehat{y}^{\mathrm{image}}$. The final fusion was

$$\widehat{y}_{ig}^{\mathrm{fusion}} = 0.8\widehat{y}_{ig}^{\mathrm{ST}} + 0.2\widetilde{y}_{ig}^{\mathrm{image}}, \tag{12}$$

where the image calibration was

$$\widetilde{y}_{ig}^{\mathrm{image}} = a_g^{\mathrm{W}} + b_g^{\mathrm{W}} \frac{\widehat{y}_{ig}^{\mathrm{image}} - a_g^{\mathrm{S}}}{b_g^{\mathrm{S}}}. \tag{13}$$

Here S denotes the two development-training patients and W the three Wu calibration patients. Moments were computed from all retained targets of the designated source and calibration patients, rather than their sparse anchor subsets or image predictions. Each patient contributed equal total weight: for a domain with n_{pat} patients and n_p locations in patient p,

$$a_g = \frac{1}{n_{\mathrm{pat}}} \sum_{p=1}^{n_{\mathrm{pat}}} \frac{1}{n_p} \sum_{i=1}^{n_p} y_{pig}, \qquad b_g = \sqrt{\frac{1}{n_{\mathrm{pat}}} \sum_{p=1}^{n_{\mathrm{pat}}} \frac{1}{n_p} \sum_{i=1}^{n_p} (y_{pig} - a_g)^2}. \tag{14}$$

A scale below 10^{-8} was replaced by 1. This scale includes variation about the shared domain center; it is not the mean of patient-specific standard deviations. All moments used the common target units in Eq. 21. Bassiouni evaluation reused these source and Wu constants without fitting new moments or fusion weights. The calibrated image term used in fusion therefore differs from the raw image-only comparator reported in Table 2.

A per-gene affine transformation $T_g(y) = (y - a_g)/s_g$ with $s_g > 0$ satisfies

$$T_g^{-1}\left(\sum_j w_{ij} T_g(y_{jg})\right) = \sum_j w_{ij} y_{jg} \quad \text{when} \quad \sum_j w_{ij} = 1. \tag{15}$$

Accordingly, the transform–interpolate–inverse-transform molecular prediction was checked as an equivalence test and omitted from duplicate primary metric tables. Floating-point differences affecting tied ranks were not counted as harmonization benefits.

4.7 Residual correction with sparse molecular graph summaries

The development comparator termed learned propagation was a gene-specific ridge residual correction using fixed graph summaries. Let H denote the equal average of the three inverse-scaled frozen image-head predictions, o the observed-location indicator and Z the expression matrix containing the permitted observed vectors and zeros elsewhere. A self-inclusive, row-normalized operator P was constructed from symmetrized six-nearest-neighbor adjacency capped at 250 μm. The observed masses and normalized molecular messages were

$$\begin{aligned} m^{(1)} &= Po, & m^{(2)} &= P^2 o, \\ V_{ig}^{(1)} &= \frac{(PZ)_{ig}}{\max(m_i^{(1)}, 10^{-6})}, & V_{ig}^{(2)} &= \frac{(P^2 Z)_{ig}}{\max(m_i^{(2)}, 10^{-6})}. \end{aligned} \tag{16}$$

These messages depend only on observed expression and coordinates. Unobserved expression values were zero in Z and were not available to the predictor.

Let X_{PCA} contain eight whitened principal components of the image embeddings. Each location had 29 shared covariates: the eight-dimensional vectors X_{PCA}, PX_{PCA} and $P^2 X_{\mathrm{PCA}}$, together with o, $m^{(1)}$, $m^{(2)}$, $\log(1 + d_1/100)$ and $n_{300}^{(16)}/16$. Here d_1 is the nearest-anchor distance in μm, and $n_{300}^{(16)}$ counts anchors within 300 μm among the up to 16 nearest observed anchors. The denominator remains 16 when fewer anchors are available. Let M denote the development-selected molecular interpolation prediction. Five additional covariates for gene g were H_g, the fixed fusion prediction $F_g = \lambda H_g + (1 - \lambda) M_g$, the zero-filled expression Z_g, and the messages $V_g^{(1)}$ and $V_g^{(2)}$. Thus each gene used 34 covariates plus an intercept.

If Φ_g denotes this design matrix with a final column of ones, the predicted expression was $\widehat{y}_g = F_g + \Phi_g \theta_g$. Coefficients minimized

$$\underset{\theta_g}{\operatorname{argmin}} \ \|(y_g - F_g) - \Phi_g \theta_g\|_2^2 + \alpha \, \|\theta_{g,1:34}\|_2^2 \,. \tag{17}$$

The intercept was unpenalized. The objective used the sum of squared residuals, without dividing by the number of fitting rows. This model had 35 coefficients per gene and 1,750 coefficients for the 50-gene panel. Fitting masks, principal-component estimation, penalty selection and controls are described in Experimental setup.

4.8 Target calibration and adaptive fusion models

The patient-weighted affine mapping in Eqs. 13 and 14 was also used to define calibration candidates for HER2ST. For source and destination centers and scales $(a_g^{\mathrm{S}}, b_g^{\mathrm{S}})$ and $(a_g^{\mathrm{D}}, b_g^{\mathrm{D}})$,

the image prediction and observed expression were expressed as

$$H^*_{ig} = \frac{H_{ig} - a^{\mathrm{S}}_g}{b^{\mathrm{S}}_g}, \qquad Y^*_{jg} = \frac{Y_{jg} - a^{\mathrm{D}}_g}{b^{\mathrm{D}}_g}. \tag{18}$$

Molecular interpolation was applied to the observed Y^*_{jg} to obtain M^*_{ig}. Predictions were subsequently mapped back by $a^{\mathrm{D}}_g + b^{\mathrm{D}}_g \widehat{Y}^*_{ig}$ for error calculation in the common target units. The unchanged-target candidate used centers zero and scales one. The mean/standard-deviation candidate used the patient-weighted moments in Eq. 14; a robust alternative replaced them with patient-weighted pooled medians and interquartile ranges. An interquartile range below 10^{-8} fell back to the corresponding standard deviation, and a standard deviation below 10^{-8} was replaced by one. Source and destination statistics were estimated separately.

The fusion candidates combined the two branches as

$$\widehat{Y}^*_{ig} = M^*_{ig} + w_i(H^*_{ig} - M^*_{ig}), \qquad 0 \le w_i \le 1. \tag{19}$$

The candidate families were molecular interpolation alone ($w_i = 0$), one global image weight, a nondecreasing three-bin rule based on nearest-anchor distance, and a five-parameter sigmoid gate. The gate used

$$\begin{aligned} w_i &= \mathrm{sigmoid}(\beta_0 + \widetilde{q}_i^{\mathsf{T}}\beta), \\ q_i &= \left[\log(1 + d_1/200),\, \log(1 + \overline{d}_{16}/200),\, \log(1 + n_{600}),\, \log(1 + v)\right]^{\mathsf{T}}, \end{aligned} \tag{20}$$

where $\widetilde{q}_i$ denotes standardized gate features, $\overline{d}_{16}$ is mean distance to the up to 16 nearest observed anchors, n_{600} is the total number of observed anchors within 600 μm, and v is the mean across genes of the unweighted population variance among the up to 16 nearest observed anchors, computed in the destination-transformed target coordinates. All distances are in μm. A single weight rule was shared across genes, regimes and budgets within each fitting fold. Patient folds, candidate grids, gate-feature standardization, optimization and selection are specified in Experimental setup. The final Wu and Bassiouni analyses retained the fixed calibrated fusion in Eq. 12; they did not fit an adaptive gate.

4.9 Experimental setup

4.9.1 Data sources and analytical roles

All experiments used previously released paired spatial expression and histology data. The four development samples were obtained through HEST[2]: NCBI783 and NCBI785 were the training patients, TENX95 was the validation patient and TENX99 was the primary development evaluation patient. The cohort comprised 7,015 training locations, 7,760 validation locations and 20,761 primary evaluation locations. Earlier conditional leave-one-patient-out analyses reused frozen gene panels and development-selected configurations and were interpreted as sensitivity analyses, not unbiased external cross-validation.

HER2ST[3] contributed one patient-specific section for each of A1–H1. Initial evaluation was followed by patient-disjoint calibration development using the same inspected cohort; these two analytical roles are distinguished throughout. In the Wu local Visium cohort[4], pa-

tients 1142243F, CID4465 and CID4535 supplied calibration statistics, and patients 1160920F, CID4290 and CID44971 remained held out. The fresh Bassiouni cohort was obtained from GSE210616[5]. The source study includes 43 sections from 22 patients. We selected the first numbered section for every published patient from metadata before target evaluation, yielding BASS_P01–P09, P11–P15 and P17–P24. Gaps in identifiers reflect source numbering rather than outcome-dependent exclusion. The source study's own reference and validation groups were not used as calibration groups in this reconstruction study.

Metadata, repository identifiers, prior-use records and available asset hashes were checked to exclude reused or reprocessed development samples. Such checks support cohort independence but cannot conclusively establish patient identity across unrelated anonymized repositories. Cohort roles and sample counts are summarized in Table 1; identifiers and exact panels are supplied in Supplementary Tables S1 and S2.

4.9.2 Expression targets and preprocessing

The primary panel contained the 50 genes ranked using development-training expression variance; ranking did not use evaluation expression. The development assay provided 280 harmonized biological genes, and larger 250- and 280-gene settings were used in earlier development analyses. The primary fresh studies retained all 50 genes. HER2ST retained the frozen 40-gene intersection without replacing missing genes. The optional 278/280 intersection in the Bassiouni metadata was not used to claim a secondary whole-transcriptome experiment.

For the second replication, the source filtered feature-barcode H5 files, tissue positions, scale-factor JSON files and tissue images were acquired for each selected section. Barcode joins and published in-tissue flags defined included locations. No new expression-derived quality threshold, tissue-class annotation or deconvolution output was used to select evaluation locations. Counts were checked for finite, nonnegative integer values. For count c_{ig} at location i and gene g, the evaluation target was

$$y_{ig} = \frac{\log(1 + c_{ig}) - \mu_g^{\mathrm{dev}}}{\sigma_g^{\mathrm{dev}}}, \tag{21}$$

where the centering and scaling constants were the pooled location mean and population standard deviation frozen from development-training patients; a standard deviation below 10^{-8} was replaced by 1. No destination library-size or capture-area correction was fitted. Identical target units were used for all methods within each comparison. These common mathematical units do not imply cross-platform calibration of absolute molecular abundance. Supplementary Fig. S2 summarizes cross-cohort patterns without pooling panels; the gene-level results are provided in Supplementary Table S3.

4.9.3 Spatial registration and image acquisition

Visium array coordinates were mapped to image coordinates by a least-squares fit to the source hexagonal lattice with 100-μm spacing. Coordinate residuals, image scale and anisotropy were checked against the frozen qualification settings, and overlays were inspected before outcome evaluation. These tests concern location consistency and do not establish cell-level registration. Images were not super-resolved.

The Wu and Bassiouni images contained only 15/25 and 24–29 native pixels, respectively, across the nominal 112-μm patch width. Resizing does not recover morphology absent from these public inputs. Image quality was therefore recorded as a limitation of the image comparator, not as a restriction on spatial interpolation that uses only coordinates and observed expression.

4.9.4 Observation masks and measurement budgets

For a section with N retained locations, the observed set $\mathcal{O}$ contained $\lfloor Nb + 0.5 \rfloor$ anchors at budget $b \in \{0.10, 0.20, 0.40\}$. Every hidden location lacked its entire expression vector; coordinates and image features remained available. Primary random trajectories used seeds 0–9 with nested prefixes, so higher budgets added anchors without discarding earlier observations. All trajectories were frozen without using expression values.

In distributed interpolation (V1), anchors were eligible throughout the tissue and all unobserved locations were scored. In contiguous regional completion (V2), a forbidden region consisted of the nearest 40% of locations to a center determined from the 25th, 50th or 75th percentile of the first coordinate principal-component rank. The three centers defined three fixed geometries. Anchors were sampled outside the forbidden region, and only the forbidden region was scored. All other unobserved labels were also removed from model inputs. Thus the same nominal budget refers to the full retained section even when evaluation is restricted to a region.

A deterministic spatial farthest-point policy was retained as a secondary sampling analysis. It started from the eligible location farthest from the eligible-coordinate centroid and selected locations without expression labels. It was not optimized on external outcomes. Earlier development block and capture-window masks differ from the final regional-completion definition and are labeled separately in Fig. 3 and the source tables.

4.9.5 Image preprocessing and frozen feature extraction

At each external location, an RGB patch covered a nominal 112-μm square, rounded to the available native pixels. The patch was resized with LANCZOS to 224×224 pixels. The torchvision `IMAGENET1K_V1` preprocessing then resized the shorter side to 256 pixels using bilinear interpolation, took a central 224×224 crop, rescaled intensities to $[0, 1]$, and normalized RGB channels using means $(0.485, 0.456, 0.406)$ and standard deviations $(0.229, 0.224, 0.225)$. Thus, 112 μm denotes the nominal extracted patch; the subsequent central crop reduces the field presented to the encoder. Development embeddings used the supplied 224×224 HEST patches with the same torchvision transform. No stain normalization, random augmentation or encoder fine-tuning was applied. Encoder parameters and batch-normalization statistics remained fixed during feature extraction.

Phikon processed the same HEST RGB patches without recropping. Its official image processor used 224×224 inputs, intensity rescaling by $1/255$ and the ImageNet channel means and standard deviations above, without stain normalization or augmentation. Extraction used evaluation mode, frozen weights, float32 precision and batches of 16.

4.9.6 Neural head fitting and model selection

Shared fitting procedure Neural heads were fitted to the 7,015 locations from NCBI783 and NCBI785, using TENX95 for validation and TENX99 for the primary development evaluation. Feature-wise means and population standard deviations were estimated from the fitting patients only. A separate per-gene scaler was fitted to their expression targets, which were already in the common development-standardized log-count units. Scales below 10^{-8} were replaced by one. For graph predictors, local image embeddings were standardized before aggregation. The additional head-target transform was inverted before scoring or averaging image predictions.

All one-hidden-layer MLPs minimized mean squared error in the additionally standardized target space using AdamW, learning rate 10^{-3}, weight decay 10^{-4}, $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\epsilon = 10^{-8}$. Weight decay applied to all head parameters, including biases. Linear-layer weights and biases used the default PyTorch initialization, uniform on $[-1/\sqrt{d_{\text{in}}}, 1/\sqrt{d_{\text{in}}}]$, where d_{in} is that layer's input width. Each epoch shuffled the fitting locations into minibatches of 512, retaining the final partial batch, for at most 80 epochs. Early stopping required validation MSE to decrease by more than 10^{-4} and stopped training after 10 consecutive epochs without such an improvement. The best qualifying parameters were restored. Dropout was disabled for validation and prediction. No learning-rate scheduler or auxiliary loss was used. Training used CPU float32 computation and deterministic PyTorch algorithms.

For the local image-head comparisons, the MLP search comprised hidden widths $\{128, 256\}$ and dropout probabilities $\{0, 0.1\}$. Hyperparameter candidates were compared using seed 0, and selected configurations were fitted with seeds 0, 1 and 2. The criterion for choosing a configuration depended on the comparison below; within-fit early stopping always used validation MSE. The three inverse-scaled predictions of the selected local image head were averaged with equal weights for subsequent sparse reconstruction. No external patient was used to select or refit these heads.

Local and representation comparisons The initial morphology-only comparison (Supplementary Fig. S3) selected the lowest-validation-MSE MLP, giving width 128 and dropout 0.1 for the 50-gene panel. Its ridge baseline used an intercept and a Cholesky solver. The penalty grid was $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100\}$; selection maximized validation median gene Pearson correlation, breaking ties by lower MSE and then smaller penalty. The selected 50-gene ridge penalty was 100. Target-row and embedding-row permutations disrupted pairing only within the fitting data, while validation and evaluation data remained unchanged.

The ResNet18–Phikon representation comparison (Supplementary Fig. S6) used the same four MLP candidates for both encoders and selected the highest validation median gene Pearson correlation, followed by lower MSE and a deterministic ordering of configurations. Both encoders selected width 256 and dropout 0.1 for 50 genes. The final ResNet18 heads retained epochs 12, 11 and 12 for seeds 0, 1 and 2, respectively. The accompanying ridge comparison used penalties $\{0.1, 1, 10, 100, 1000\}$, with 1000 selected for both encoders. The equal search budget and shared hidden width did not equalize the number of parameters because ResNet18 and Phikon supplied 512- and 768-dimensional inputs, respectively. The width-128 initial MLP and the width-256 representation-comparison MLP were distinct fitted predictors.

Spatial aggregation and topology controls The spatial graph candidates defined in Methods were compared at seed 0 by validation median gene Pearson correlation, followed by lower MSE and deterministic graph-name order. All three 50-gene spatial predictors selected Delaunay adjacency capped at 250 μm. Mean pooling and GCN used hidden width 128; mean-GraphSAGE used width 67 to approximately match the local predictor's parameter count after concatenating local and neighbor features. All used dropout 0.1 and the shared neural fitting procedure.

The matched topology controls (Supplementary Fig. S5) retained the selected mean-GraphSAGE head. Degree-preserving double-edge rewiring targeted five accepted swaps per edge, with at most 100 attempts per edge. The distance-constrained variant also preserved edge counts in $[0, 175)$, $[175, 250)$ and $[250, \infty)$ μm, used candidate endpoints within 275 μm and allowed at most 1,000 attempts per edge. Five graph randomizations were crossed with the three training seeds. Model fitting and validation early stopping were performed separately for each graph and training seed. The Phikon spatial comparison inherited the selected graphs, hidden widths and dropout: width 128 for mean pooling and width 67 for mean-GraphSAGE. True spatial graphs were evaluated with three training seeds, and degree-plus-length-matched controls used five graph randomizations crossed with those seeds. These inherited widths did not provide exact parameter matching across encoders.

Conditional development sensitivity fits Conditional leave-one-patient-out fits used the other three development patients, refitted feature and head-target scalers on those patients, and retained the primary-selected configurations. Training used fixed epoch counts without early stopping on the held-out patient. For 50 genes, these counts were 14 for the initial local MLP, 14 for mean pooling and GCN, 7 for mean-GraphSAGE, 12 for the representation-comparison ResNet18 head and 1 for the Phikon head. The counts were rounded medians of the corresponding primary fits' best epochs. The gene panel and underlying development target definition were unchanged, so these analyses describe sensitivity conditional on prior development choices.

4.9.7 Sparse reconstruction fitting and controls

The residual correction model used image-feature standardization and eight-component whitened principal-component analysis fitted to the training patients only. Principal components were obtained by randomized singular-value decomposition with random seed 0. No additional expression-derived scaler was fitted. For each training mask, up to 256 hidden locations were sampled without replacement using that mask's seed. Their expression entered the residual-regression response only; molecular features were constructed from the observed anchors. The fitting masks covered 5, 10, 20, 40, 60 and 80% of locations under random, block and capture-window sampling, with three replicates per condition.

Mean interpolation selected its neighborhood size from $\{1, 4, 8, 16\}$; inverse-distance interpolation used up to 16 anchors and selected its power from $\{1, 2, 3\}$. The stronger interpolation family and fusion image weight $\lambda \in \{0, 0.25, 0.5, 0.75, 1\}$ were selected by mean validation MSE separately by panel and sampling geometry, with equal weights for masks. For the 50-gene panel, random sampling selected inverse-distance-squared interpolation and $\lambda = 0.25$, block sampling selected the 16-anchor mean and $\lambda = 0.5$, and capture-window sampling

selected inverse-distance-cubed interpolation and $\lambda = 0.5$. These settings describe the development comparisons; the final external protocol used the separately specified interpolation and fusion rules.

After interpolation and fusion were selected, the residual ridge penalty was chosen from $\{1, 100, 10000\}$ by mean validation MSE across all masks, with equal mask weights. Both encoders selected $\alpha = 10000$ for 50 genes. The unpenalized-intercept, sum-of-squares objective is given in Eq. 17. The conditional patient analyses refitted image standardization, principal components and residual coefficients on the other three patients while retaining the primary-selected interpolation, fusion and regularization settings; they reused the corresponding frozen conditional image predictions.

The controls in Supplementary Fig. S7 used TENX99, the 50-gene panel and one random 20% observation mask (replicate 0). Three permutation seeds reassigned observed expression vectors, anchor positions or image features and predictions while leaving the fitted models unchanged. In this figure, B1, B3 and B4 denote the frozen image predictor, fixed fusion and ridge residual correction, respectively. The two anchor-assignment controls disrupt related expression–location associations and do not represent independent biological mechanisms.

4.9.8 HER2ST calibration and fusion selection

The target-mapping and selected-fusion analyses used the same eight HER2ST patients and 40 shared genes (Supplementary Figs. S1 and S10). Four outer folds held out the patient pairs A1/B1, C1/D1, E1/F1 and G1/H1, respectively. Within each remaining six-patient set, target mappings were assessed using leave-one-patient-out records: the mapping for an inner patient's predictions was estimated from the other five patients. Source-domain moments were fixed using NCBI783 and NCBI785. The candidate mappings were the unchanged target, per-gene mean/standard-deviation mapping and per-gene median/interquartile-range mapping. Source and destination statistics were fitted separately, with equal total weight per patient.

Target-mapping selection retained fixed image weights 0.25 for V1 and 0.5 for V2. Inner mask seeds were 0–2. Four conditions received equal weight: V1 coverage 10, 20 and 40%, and V2 coverage 20%. Seeds and geometries were averaged within patient and condition, then conditions within patient, then the six patients. Candidates within 0.005 of the highest mean patient median gene Pearson correlation were eligible; among these, selection minimized MSE in the common target units. Every fold selected the mean/standard-deviation mapping. The corresponding outer mapping used destination moments estimated from all six outer-training patients.

Fusion selection used the resulting inner records to compare interpolation alone, a global image weight from $\{0, 0.1, \ldots, 1\}$, 35 nondecreasing three-bin image-weight rules, and the sigmoid gate defined in Eq. 20. The bin rules used nearest-anchor intervals $[0, 300)$, $[300, 600)$ and $[600, \infty)$ μm and all nondecreasing triples from $\{0, 0.1, 0.25, 0.5, 1\}$. Each candidate used a single rule shared across genes, regimes and budgets within a fold.

The sigmoid gate minimized common-unit squared error with equal contributions from patients, conditions, runs and locations before division by each run's interpolation MSE, floored at 10^{-8}. The objective also included slope penalty $0.01\|\beta\|_2^2$ and intercept penalty $10^{-6}\beta_0^2$. The same observation weights defined the gate-feature means and population standard devi-

ations; standard deviations below 10^{-8} were replaced by one. L-BFGS-B used initial coefficients $(-2, 0, 0, 0, 0)$, bounds $[-12, 12]$, at most 300 iterations, function tolerance 10^{-12} and gradient tolerance 10^{-7}. Gate fitting and candidate selection used the inner records from the outer-training patients; no outer-patient outcome entered either operation.

The correlation-then-MSE criterion above also selected the fusion family. The first three patient-pair folds selected global image weight 0.2; the G1/H1 fold selected the sigmoid gate. Outer evaluation used mask seeds 0–9. Supplementary Fig. S10 reports these fold-selected rules, whereas Supplementary Fig. S1 reports the fixed-fusion target-mapping comparison. Both analyses remain calibration development within previously examined HER2ST patients.

4.9.9 Metrics, patient aggregation and decision criteria

All expression metrics were computed on hidden locations only; the full observed expression vector was available at each anchor and the full target vector was withheld at every other location. Observed rows were selected before molecular prediction, so hidden expression could not influence anchor values or interpolation weights. For each gene, Pearson and Spearman correlations summarized spatial association, while MSE and MAE summarized prediction error. Panel Pearson values were medians across genes with defined correlations. Correlations were undefined when the target or prediction range was at most 10^{-12}; Spearman correlations used average ranks for ties. Undefined correlations were excluded from panel summaries; such genes did not pass positive-correlation threshold counts. Panel MSE and MAE were means of per-gene errors. In formulae, for the scored hidden set $\mathcal{H}$,

$$\mathrm{MSE} = \frac{1}{G|\mathcal{H}|}\sum_{g=1}^{G}\sum_{i\in\mathcal{H}}(\widehat{y}_{ig} - y_{ig})^2, \qquad \mathrm{MAE} = \frac{1}{G|\mathcal{H}|}\sum_{g=1}^{G}\sum_{i\in\mathcal{H}}|\widehat{y}_{ig} - y_{ig}|. \tag{22}$$

Within each patient and geometry, metrics were averaged over mask seeds; geometry means were then averaged equally. Headline cohort values were medians of these patient summaries. Tables label an MSE value as a cohort median of patient mean MSEs, not a pooled mean over all spots. Paired effects were calculated within patients before taking their median. This distinction is necessary because a median paired effect need not equal the difference or ratio of marginal medians.

The patient was the biological unit. Seeds described computational or mask variability, not additional patients. We report descriptive effects and patient-support counts without new hypothesis tests, spot-level significance claims or post-hoc population confidence intervals. The second-replication support criterion required a median gene Pearson gain of at least 0.02 over image-only prediction, relative mean MSE reduction of at least 2%, and non-worsening MAE. The cohort rule required at least $\lceil 2N/3 \rceil$ patients meeting the joint criterion at at least two budgets, positive median benefits at every budget, and the same support threshold for correlation/MSE improvement from 10% to 40%. For 22 patients this threshold was 15. The limited native image resolution restricts interpretation of the image comparisons even when the molecular support criterion is satisfied.

4.9.10 Distance-resolved evaluation

For each hidden location, the nearest-anchor distance was $d_i = \min_{j \in \mathcal{O}} \|p_i - p_j\|$. Distances were grouped into the frozen bins $[0, 150)$, $[150, 300)$, $[300, 600)$, $[600, 1200)$, $[1200, 2400)$ and $[2400, \infty)$ μm. Error summaries were reported for occupied bins; the Bassiouni binwise correlations required at least 20 scored locations, while the Wu distance summaries contained errors only. Empty bins were omitted, not filled with zero. Counts, geometry and run identities were retained to distinguish sparse support from low error. Nearest-versus-farthest contrasts and monotonicity summaries were descriptive properties of the occupied-bin curves, not estimates of a universal biological length scale.

4.9.11 External negative controls and numerical verification

In the second replication, control patient BASS_P01, 20% coverage, random-mask seed 0, V1 geometry 0 and V2 geometry 1 were designated before evaluation. Three control seeds permuted observed expression-vector assignments, anchor-position assignments or image predictions. The first two disrupt related molecular–location associations; they were not interpreted as independent biological evidence. Every saved prediction was also repeated after all unobserved expression was replaced with NaNs and with independent random values. Observed/hidden barcodes, mask identities, configurations and hashes accompanied the predictions.

4.9.12 Software

Analyses were implemented in Python using PyTorch, NumPy[21] and SciPy[16]. The code repository provides the analysis implementation and dependency information. The accompanying `methods_data/` files provide the model settings, full-precision image-feature scaler, target-normalization constants and source-to-Wu calibration constants.

4.10 Ethics

This work reanalyzed publicly released, anonymized datasets and involved no new recruitment, tissue collection or experimental intervention. The original studies describe their ethics and consent procedures[3–5].

5 Data availability

The development data were drawn from HEST[2]. HER2ST data are available at https://zenodo.org/records/4751624; the Wu local spatial dataset at https://zenodo.org/records/4739739; and the Bassiouni spatial dataset under GEO accession GSE210616, https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE210616. The accompanying source-data files contain the tabulated results underlying the manuscript and supplementary figures. The original public datasets remain available from their source repositories. Raw tissue images and complete prediction caches are not redistributed in the manuscript package.

6 Code availability

The analysis code, model definitions and evaluation scripts are publicly available at https://github.com/HySonLab/GraphPathology. Numerical outputs supporting the figures and tables are included with the accompanying source data.

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No funding to disclose.

9 Author contributions

Truong-Son Hy was responsible for study conception and design, methodology, computational analysis and validation, interpretation, visualization, and manuscript writing and revision.

10 Competing interests

Truong-Son Hy is an Editorial Board Member of *Scientific Reports*. The author declares no other competing interests.

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